

Assignment 1

Deployment Environment and Cloud Platform Recommendation

Use case: FraudGuard - synthetic mobile-money fraud detection

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Chosen cloud platform: Amazon Web Services (AWS)

Project repository: https://github.com/AradhyaMarya047/INFT4100_project

AI-use statement: Generative AI was used to assist with structuring, drafting, and understanding the tasks for the assignment. Project-specific claims are based on the work done in class till now and assignment materials. The scores, recommendations, and cost assumptions are based on the research and logic.

1. Use case and readiness snapshot

1.1 Use case (under 200 words)

FraudGuard is a machine-learning fraud-detection prototype for a synthetic mobile-money platform. The intended users are a transaction-processing application and fraud operations staff. For each incoming transaction, the service would use transaction type, amount, origin/destination balance features, and related fields to estimate whether the transaction is fraudulent; the application could then allow, flag, or route the transaction for review. The repository's Week 1/Week 2 work uses a PaySim-style synthetic dataset with 200,000 records, five transaction types, an isFraud target, and a legacy isFlaggedFraud indicator. The current baseline is a scikit-learn Random Forest classifier. The most important lifecycle stages after experimentation are independent model evaluation, deployment, monitoring for data/model drift, and controlled retraining. The dataset is synthetic and the repository demonstrates a prototype, not a validated financial product; real customer data and live decisions are outside the current evidence base.

1.2 Why production AI is needed

Fraud patterns can change and decisions may be needed while a transaction is being processed, so a notebook-only model cannot provide dependable service, operational monitoring, or an auditable decision path. In a real deployment, missed fraud could cause financial loss, while false alarms could block legitimate customers; availability, latency, security, and rollback therefore matter as much as offline model performance.

1.3 Readiness scorecard

The project is moderately ready for further engineering but is not ready to launch. The six scores below follow the course scorecard's 0–5 scale; each has one evidence sentence and a gap classification. The scores are judgments based on repository evidence, not measured production service levels.

Data	Model	Infrastructure	Organizational	Risk	Cost
3	2	1	2	2	3
Fix before launch	Fix before launch	Fix before launch	Fix before launch	Fix before launch	Monitor after launch

1.4 Two weakest dimensions and deployment effect

Infrastructure (2/5) is weak because the repository currently centres on notebooks, a local dataset, a generator, dependencies, and a saved baseline rather than an operational service with monitored serving, rollback, recovery, and deployment automation. This favours managed AWS services and a staged pilot rather than self-managed production infrastructure. Risk (2/5) is also weak because fraud decisions have asymmetric consequences and the current prototype does not demonstrate production controls for security, false-positive/negative thresholds, drift, audit, or incident response. These are launch blockers; cloud hosting alone does not make the model safe.

Readiness summary (Appendix A contains the completed table): total 17/30; average 2.83/5; weakest dimensions are Infrastructure and Risk (2/5 each). Decision: Delay for

Remediation.

2. Deployment environment evaluation

2.1 Environment Fit Scorecard

The scorecard is applied in the required order: data sensitivity, cost pattern, operational capacity, then latency. The use case is explicitly a synthetic-data prototype today; the proposed future production system would process sensitive transaction data, so the score reflects the proposed workload and assumes that required controls and contractual/regulatory reviews permit the data to be processed in the selected AWS Region.

Data sensitivity	Cost pattern	Operational capacity	Latency	Average	Spread
3	4	5	3	3.75	2

2.2 Three-environment comparison

On-premises gives the strongest direct physical control but would require purchasing, securing, patching, scaling, and recovering the inference platform. Cloud provides managed services, elastic capacity, and consumption-based billing, but requires correct configuration and ongoing customer security responsibility. Hybrid can keep a regulated source system or final approval decision on-premises while using cloud for de-identified training; however, it adds integration and operational complexity. For this use case, cloud is the best initial target because the current project has no demonstrated 24/7 operations team and demand is assumed to be variable.

At a glance	On-premises	Cloud (AWS)	Hybrid
3-year cost	about \$395k	about \$104k	about \$220k (Week 2 model)
People to run it	about 0.75 FTE	about 0.25 FTE	more than either alone
Data control	Full	Shared, in Canada	Strongest per field
Fit for FraudGuard	Only if data must stay on-site	Recommended	Fallback (tipping point)

2.3 Recommendation

Recommendation: use AWS public cloud for the proposed FraudGuard pilot, subject to a security and data-governance review before any real transaction data is introduced. The decisive scorecard question is Operational Capacity: the project does not demonstrate an in-house team able to run infrastructure around the clock, so managed services reduce the amount of hardware operations required. Cost Pattern also supports cloud because a small pilot can scale with demand rather than buying hardware sized for peak volume. Data Sensitivity is scored 4 rather than 5 because a real financial deployment would need explicit approval for data location, access, retention, and applicable legal obligations; cloud is not automatically permitted just because AWS has compliance documentation. Latency is scored 4 because near-real-time scoring is the intended direction, but no measured service-level objective has yet been established. The scorecard rule agrees with cloud: no data/latency hard-limit score of 1, spread is

below 3, and average is at least 3.6.

2.4 Three-year cost reasoning

The estimates are order-of-magnitude planning figures in Canadian dollars, not quotes. Assumption: a pilot averaging up to 200,000 scoring requests per month, a small always-on inference service, 100 GB of stored data/artifacts/logs, limited model retraining, one AWS Region, and no multi-region disaster recovery. The repository's 200,000 rows are a teaching dataset, not evidence of real monthly traffic; 200,000 requests/month is a planning scenario only.

Cost line (3 years)	Option A: AWS cloud	Option B: on-premises	Uncertainty
Compute (serving, training, load balancer)	\$4.2k	\$86k (includes hardware)	Medium
Storage and data transfer	\$0.6k	\$15k	Low
Managed services, licenses, support	\$9.0k	\$24k	Medium
Staff time to operate	\$90k	\$270k	High
Three-year total (order of magnitude)	about \$104k	about \$395k	

2.5 Tipping point

Reconsider the cloud-only recommendation if a regulator, customer contract, or security review requires raw transaction data and inference to remain inside an organisation-controlled network. In that case, a hybrid design could keep raw data and time-critical decisions on-premises while sending only approved, de-identified training data to AWS.

3. Cloud platform and service selection

3.1 Platform choice

Amazon Web Services (AWS). First, the proposed service map covers managed storage, API ingress, serverless request validation, model training/hosting, monitoring, and audit without requiring a self-managed data centre. Second, AWS offers a mature managed machine-learning service and documented patterns for fraud-detection workloads.

Third,

pay-as-you-go resources fit a small pilot with uncertain demand, provided budgets and alerts are established. Fourth, AWS's Cloud Adoption Framework and Well-Architected guidance give a structured path for governance, security, reliability, and cost management. Azure is not selected because no existing Azure dependency or project-specific advantage has been established in the evidence. Google Cloud is not selected for the same reason: the project has no demonstrated GCP dependency that outweighs the benefit of standardising on one platform. This is a project-fit decision, not a claim that either alternative is technically inferior.

3.2 Service map

The proposed service map is in Appendix C. Amazon S3 stores versioned datasets and approved model artifacts; Amazon SageMaker AI supports training, evaluation, and managed inference; Amazon API Gateway exposes an HTTPS endpoint; AWS

Lambda validates and transforms incoming requests; Amazon ECR stores container images where a custom inference image is required; Amazon CloudWatch collects logs, metrics, and alarms; AWS Identity and Access Management (IAM) restricts permissions; and AWS CloudTrail records account/API activity. These are all AWS services. The architecture is a proposal and none of these services is claimed to be deployed in the current repository.

Building block	Named service	Model	Deployment style
Training compute	Amazon SageMaker AI training jobs	PaaS	Managed container
Serving / hosting	Amazon ECS on AWS Fargate behind an Application Load Balancer	PaaS	Container
Data storage	Amazon S3 with AWS KMS encryption	IaaS	Managed storage
Managed AI / ML	Amazon SageMaker AI (Model Registry, Model Monitor, Clarify)	PaaS	Managed
Alerting	Amazon SQS, AWS Lambda, Amazon SNS	PaaS	Serverless
Network and security	Amazon VPC, AWS Site-to-Site VPN, IAM, KMS, CloudTrail, CloudWatch	IaaS / PaaS	Managed

3.3 Containers vs. serverless

For the request-validation/API-adapter step, choose serverless AWS Lambda behind Amazon API Gateway. The assumed traffic is variable, the validation function should be short-lived, and the code is small enough that serverless reduces idle-compute management. For the Random Forest model-serving component, use Amazon SageMaker AI real-time inference for the pilot so model hosting and endpoint operations are managed; if custom packaging or portability requirements dominate later, package the scikit-learn model as a container and evaluate deployment on Amazon ECS with AWS Fargate. The choice should be validated with a load test because the actual latency and package-size characteristics have not been measured. Training is an intermittent job and can use managed SageMaker AI training rather than an always-on training server.

3.4 Build vs. buy

Recommendation: use a managed AI/ML service for hosting and lifecycle scaffolding, while retaining the project's own Random Forest model. A fully self-managed stack offers more control over dependencies and serving internals, but requires the team to operate servers, patch runtimes, build deployment automation, and implement monitoring and rollback.

SageMaker AI reduces some operational work and supports training and hosting workflows, improving time to value for a small team. The trade-off is service-specific configuration, possible higher costs at low utilisation, and lock-in around model deployment interfaces. The model itself should remain a portable scikit-learn artifact with a documented feature schema and repeatable evaluation tests.

Criterion	Managed AI service (Amazon Fraud Detector)	Managed platform, own model (SageMaker AI)	Self-managed (EC2 or EKS with MLflow)
Time to value	Fastest, but closed to new customers since 7 Nov 2025	Days: my scikit-learn code runs in managed training	Weeks: build pipelines, monitoring, patching
Control and customization	Low: fixed model and features	High: my features, thresholds and reason codes	Highest: any framework or hardware codes
Cost over time	Not priced; unavailable	Pay per hour; about \$1.4k a year compute at pilot scale	Lowest unit price, highest staff time
Expertise required	Low	Medium: one part-time ML and DevOps owner	High: platform and ML-operations engineers

3.5 Cloud Adoption Framework

Plan: define business outcomes, a pilot budget, transaction-volume assumptions, latency target, fraud thresholds, data classification, and a clear stop/go gate. Ready: create separate development and production accounts or equivalent account boundaries; configure IAM least privilege, MFA, encryption, audit logging, approved Region, secrets handling, budget alerts, and a repeatable deployment environment. Adopt: deploy a small pilot using synthetic data first; version the model and data; run load, failure, security, and model-quality tests; release through a controlled pipeline with rollback. Govern & Operate: monitor API availability, latency, errors, cost, data quality, drift, precision/recall, false-positive rate, and false-negative rate; review access and audit logs; rehearse incidents; and require approval before a model is promoted. These steps apply AWS CAF's governance, security, platform, people, and operations capabilities to this project [8].

3.6 Vendor lock-in

The largest lock-in points are SageMaker endpoint configuration and model lifecycle integration, API Gateway/Lambda event formats, IAM policy structure, and CloudWatch dashboards/alarms. Mitigation: keep the model artifact and preprocessing code independent of AWS SDK calls; define a stable HTTP request/response schema; version feature transformations; store data in open formats such as CSV or Parquet in S3; manage infrastructure as code; and test that the container can run locally. Portability is not free, but these choices make migration or a hybrid design less expensive.

4. Risks and decision

4.1 Top three risks

Risk 1: - Incorrect fraud predictions (likelihood: High; impact: High). False negatives can allow fraud, while false positives can interrupt legitimate transactions. Mitigation: set business-approved thresholds using precision-recall analysis; evaluate by transaction type and time period; use a human review path for uncertain scores; monitor outcomes; and do not automatically block real customers until validation and approval are complete.

Risk 2: - Security or data exposure due to misconfiguration (likelihood: Medium; impact: High). AWS follows a shared-responsibility model: AWS secures the underlying cloud, but the customer

remains responsible for appropriate data, identity, network, and service configuration [10]. Mitigation: least-privilege IAM roles, MFA, encryption in transit and at rest, private networking where appropriate, secrets management, CloudTrail, log redaction, vulnerability checks, and a documented incident-response plan.

Risk 3: - Cost growth or service degradation under load (likelihood: Medium; impact: Medium–High). Unbounded requests, excessive logging, an oversized always-on endpoint, or untested spikes could increase cost or degrade latency. Mitigation: AWS Budgets and alerts, service quotas, request limits, load tests, CloudWatch alarms, right-sizing, and a cost review before scaling. This reflects the cost-optimization and reliability trade-offs highlighted by AWS Well-Architected.

4.2 Go / Delay / No-Go

Decision: DELAY FOR REMEDIATION. The readiness score is 17/30 (average 2.83/5), and the scorecard guidance indicates that readiness is still developing below 3.5. The environment-fit score supports AWS as the target environment, but that does not override the readiness gaps. A GO decision requires all of the following: (1) independent evaluation on held-out and time-separated data with approved precision/recall and false-positive/negative thresholds; (2) validated API/model serving with load, failure, recovery, and rollback tests; (3) documented data classification, access controls, encryption, audit logging, and incident response; (4) monitoring for input quality and drift; (5) an approved three-year cost estimate and budget alerts; and (6) an accountable owner and operational runbook. Until these are met, use synthetic data only.

5. Communication and professional practice

5.1 Architecture diagram

Figure 1 presents the proposed AWS boundary, named services, request/response path, model training path, and monitoring. The design assumes HTTPS ingress, an API validation step, managed inference, and a separate training/evaluation flow. It is a target-state diagram, not evidence of a deployed system. The diagram's security baseline is least-privilege IAM, encryption, audit events, and restricted network access.

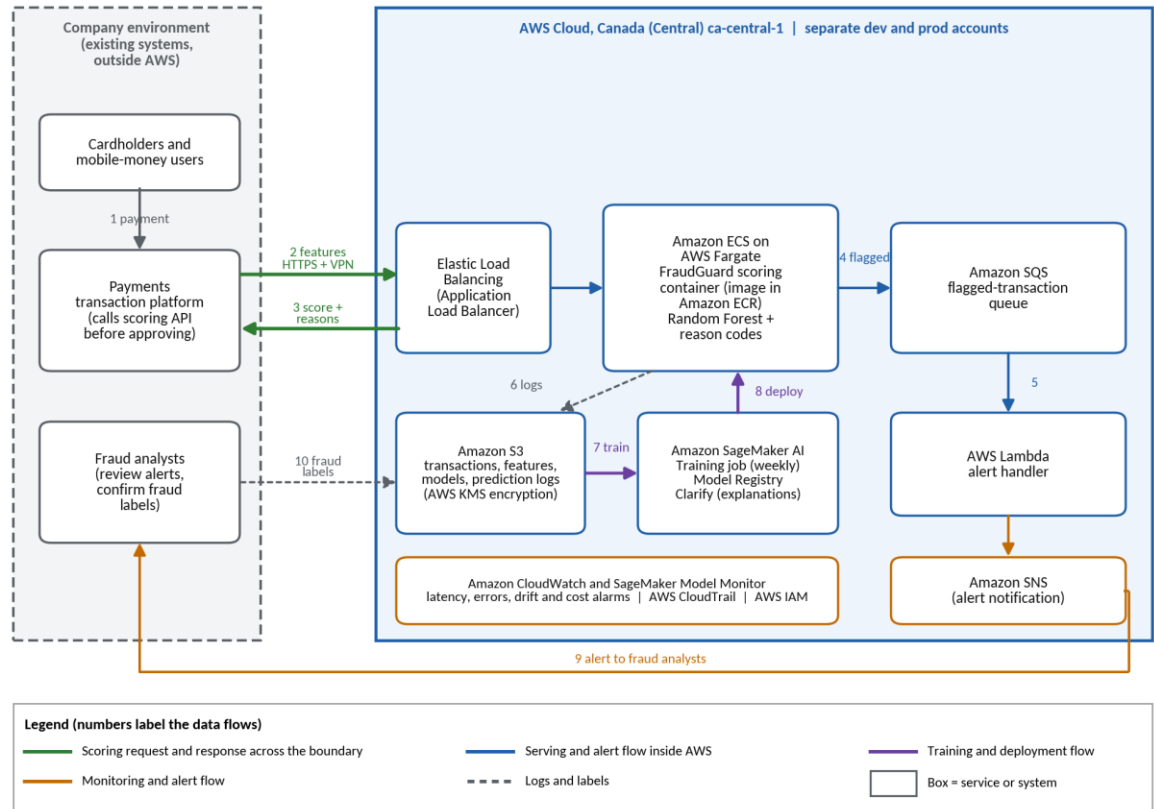


Figure 1. FraudGuard deployment diagram (company environment and AWS Canada (Central)).

5.2 Version control and project continuity

The report continues the existing course repository, which contains Week01.ipynb, Week02.ipynb, the shared synthetic data generator, the generated Week 1 dataset, and the requirements file. Repository URL: https://github.com/AradhyaMarya047/INFT4100_project. Committed to the same repository in the folder docs.

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Appendix A: - Completed Readiness Scorecard

Dimension	Score (0-5)	Evidence (one specific sentence)	Gap type and effect on the deployment decision
Data	3	200,000 synthetic rows with 0 duplicates and 0.0185% missing values are structurally clean, but all are synthetic at a 0.6% fraud rate (real PaySim-style data is about 0.13%), so no real fraud pattern or drift has been tested.	Fix before launch. Four weeks in shadow mode on real, de-identified data; store data in S3 in ca-central-1 with KMS keys.
Model	2	On fresh unseen data (seed 123) the Random Forest reached fraud precision 0.92 and recall 0.86, but it has no tuned threshold, no explanation for a block, and no test on drifted data.	Fix before launch. Needs Clarify reason codes, threshold tuning, drift tests and a registry with rollback, so training and registry go on a managed ML platform.
Infrastructure	1	No deployment environment exists: the model is a saved joblib file and notebooks, with no API, load test, monitoring or on-call process.	Fix before launch. Hard blocker. Drives the choice of cloud (days to stand up) and the landing-zone condition in Section 4.2.
Organizational	2	IT has general cloud experience but has never run an ML model in production, and nobody owns on-call or retraining.	Fix before launch. Pushes to managed services (Fargate, SageMaker AI) over Kubernetes or own servers; name a part-time ML and DevOps owner and write runbooks.
Risk	2	Compliance requires an explainable reason for every blocked transaction and none is produced; a wrong block stops a customer's payment and a missed fraud loses money.	Fix before launch. Requires reason codes, audit logs and shadow mode before enforcement; keeps identifiable data minimal.
Cost	3	Finance approved a 3-month pilot budget only; my estimate is about \$104k over 3 years on AWS, but no production budget exists.	Monitor after launch. Favours pay-per-use services; budget alerts at 50/80/100% of the pilot cap; revisit before year 2.

Production Readiness Scorecard

FraudGuard

Aradhya

08-10-2026



1. Data

Is our data trustworthy enough to run a business decision on?

- 1 No validation, unknown lineage
- 3 Validated, but lineage undocumented
- 5 Validated, versioned, governed

0 1 2 **3** 4 5

EVIDENCE, ONE SENTENCE

200,000 synthetic rows with 0 duplicates and 0.0185% missing values are structurally clean, but all are synthetic at a 0.6% fraud rate (real PaySim-style data)

- ☒ Fix before launch
 ☐ Monitor after launch
 ☐ No gap

2. Model

Does the model perform reliably outside training conditions?

- 1 Never tested outside training data
- 3 Tested once, no bias/drift plan
- 5 Tested, explainable, monitored for drift

0 1 **2** 3 4 5

EVIDENCE, ONE SENTENCE

On fresh unseen data (seed 123) the Random Forest reached fraud precision 0.92 and recall 0.86, but it has no tuned threshold, no explanation for a block, and

- ☒ Fix before launch
 ☐ Monitor after launch
 ☐ No gap

3. Infrastructure

Can our systems handle real load, failure, and change?

- 1 Runs on someone's laptop
- 3 Deployed, but no monitoring/rollback
- 5 Containerized, monitored, autoscaled, recoverable

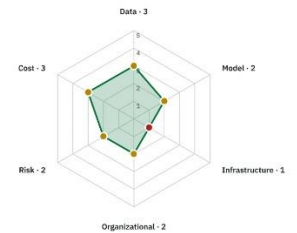
0 **1** 2 3 4 5

EVIDENCE, ONE SENTENCE

No deployment environment exists: the model is a saved joblib file and notebooks, with no API, load test,

- ☒ Fix before launch
 ☐ Monitor after launch
 ☐ No gap

READINESS RADAR



TOTAL **13** / 30
 AVERAGE **2.2** / 5
 LOWEST **Infrastructure**

4. Organizational

Do we have the people and process to run this responsibly?

- 1 No named owner
- 3 Owner named, no documented process
- 5 Owner, on-call, governance all documented

0 1 **2** 3 4 5

EVIDENCE, ONE SENTENCE

IT has general cloud experience but has never run an ML model in production, and nobody owns on-call or retraining.

- ☒ Fix before launch
 ☐ Monitor after launch
 ☐ No gap

5. Risk

What's the worst case if this goes wrong, and can we live with it?

- 1 No risk review done
- 3 Risks identified, not mitigated
- 5 Risks identified, mitigated, and monitored

0 1 **2** 3 4 5

EVIDENCE, ONE SENTENCE

Compliance requires an explainable reason for every blocked transaction and none is produced; a wrong block stops a customer's payment and a missed fraud

- ☒ Fix before launch
 ☐ Monitor after launch
 ☐ No gap

6. Cost

Does the value this creates outweigh what it costs to keep alive?

- 1 Cost never estimated
- 3 Cost estimated, not compared to value
- 5 Cost tracked and clearly justified by value

0 1 2 **3** 4 5

EVIDENCE, ONE SENTENCE

Finance approved a 3-month pilot budget only; my estimate is about \$104k over 3 years on AWS, but no production budget

- ☐ Fix before launch
 ☒ Monitor after launch
 ☐ No gap

GO / NO-GO

Red flag: a score of 0 or 1 needs a leadership conversation before launch.

- ☐ Ship (go)
 ☒ Delay for remediation
 ☐ Kill (no-go)

CONDITIONS AND NEXT ACTIONS (OWNER AND DATE)

Complete independent model evaluation; validate production serving, failure recovery and rollback;

REVIEWED BY (NAME, ROLE, DATE)

Student self-assessment; instructor/peer review

© Copyright Omar Altrad • INFT 41000 Readiness Scorecard. A ready-to-ship use case does not need a 5 everywhere, but a 1 anywhere is a conversation leadership must have first.

Appendix B1: - Environment Fit Scorecard

Question (in order)	Facts about my use case	Score (1 = on-premises, 5 = cloud)
Data sensitivity	Production data would be regulated payment records (account IDs, balances), but the pilot uses synthetic data, records can be tokenised, and AWS Canada (Central) keeps data in Canada; nothing forces on-premises.	3
Cost pattern	50,000 transactions a day with 10-times holiday peaks, uncertain growth and bursty weekly retraining; pay-per-use beats buying for the peak (Week 2 model: \$88.6k cloud against \$281k on-premises at pilot volume).	4
Operational capacity	IT has cloud experience but no ML-operations staff or on-call owner; a managed platform avoids hiring for hardware, patching and capacity.	5
Latency	The score must return before a payment is approved (my target: p99 under 100 ms); a small tabular model and same-region calls fit, but a network hop leaves less headroom than an on-site server.	3

Environment Fit Scorecard

USE CASE

FraudGuard

PREPARED BY

Aradhya

DATE

08-10-2026



1. Data sensitivity

Can this workload's data leave our own network?

- 1 Regulated or highly sensitive; cannot leave our network
- 3 Mixed: some regulated data, some not
- 5 Low sensitivity or de-identified; provider certifications suffice

1

2

3

4

5

EVIDENCE, ONE SENTENCE

Production data would be regulated payment records (account IDs, balances), but the pilot uses synthetic data, records can be tokenised, and AWS Canada (Central) keeps data in Canada; nothing forces on-premises.

2. Cost pattern

Is demand steady enough to justify owning hardware, or bursty enough to rent it?

- 1 Steady, predictable, high volume; owning hardware pays off
- 3 Steady baseline with occasional peaks
- 5 Unpredictable or bursty; pay-as-you-go beats idle hardware

1

2

3

4

5

EVIDENCE, ONE SENTENCE

50,000 transactions a day with 10-times holiday peaks, uncertain growth and bursty weekly retraining; pay-per-use beats buying for the peak

3. Operational capacity

Do we have the in-house team to run this 24/7, and do we want to?

- 1 Strong in-house team can run it around the clock
- 3 Some skills, gaps in on-call or specialist roles
- 5 No in-house team; the provider must manage the hardware

1

2

3

4

5

EVIDENCE, ONE SENTENCE

IT has cloud experience but no ML-operations staff or on-call owner; a managed platform avoids hiring for hardware, patching and capacity.

4. Latency

How much delay can this system tolerate between data and compute?

- 1 Near-zero tolerance; must run beside the data or on site
- 3 Moderate tolerance; some network delay is acceptable
- 5 Delay-tolerant; batch or asynchronous work is fine

1

2

3

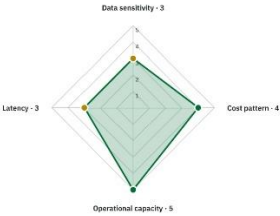
4

5

EVIDENCE, ONE SENTENCE

The score must return before a payment is approved (my target: p99 under 100 ms); a small tabular model and same-region calls fit, but a network hop leaves less headroom than an on-site server.

FIT PROFILE



CENTRE: ON-PREMISE

EDGE: CLOUD

AVERAGE

3.8

/ 5

SPREAD

2

HIGH MINUS
LOW

HARD LIMITS

None

RECOMMENDATION

Cloud candidate. Average 3.8: elastic demand and low constraints point to cloud.

☐ On-premise ☒ Cloud ☐ Hybrid

ONE-SENTENCE JUSTIFICATION

AWS public cloud for a controlled pilot using synthetic data, subject to security/data-

QUESTION THAT DROVE THE DECISION

Operational capacity



Rule, in order: a 1 on data or latency rules out public cloud alone. A spread of 3 or more means hybrid. Otherwise 2.4 or less is on-premise, 3.6 or more is cloud, and in between operational capacity breaks the tie.

Appendix B2: - Environment comparison for FraudGuard

Dimension	On-premises	Cloud (AWS)	Hybrid
Data control	Full: transaction records never leave company facilities; the company patches, encrypts and logs the servers that hold them.	Shared: data stay in Canada (Central), encrypted with company-held KMS keys; AWS secures the facilities, the company configures access.	Strongest per field: identifiable data and live scoring stay on-site, only de-identified features go to the cloud; two boundaries to secure.
Cost model	About \$50k capital (2 CPU servers, network) plus about \$25k a year hosting, software and backup, sized for the holiday peak; about \$395k over 3 years.	Pay per use: about \$1.4k a year compute and \$3k a year managed services; about \$104k over 3 years.	Both: on-site scoring servers plus cloud training and a VPN; Week 2 model put it at \$220k against \$89k for cloud at pilot volume.
Scalability	Fixed: two servers handle today's peak of about 23 requests a second, but 100 times growth or heavier models mean buying hardware with weeks of lead time.	Elastic: Fargate adds tasks in minutes for holiday spikes, and weekly training rents compute only while it runs.	Cloud side scales; the on-site scoring tier does not, so holiday peaks cap at owned servers unless cloud overflow is built.
Operational burden	High: hardware, OS patching, network, backups and on-call, about 0.75 FTE, for a team that has never run ML in production.	Low to medium: AWS runs hardware and the ML platform; the team owns configuration, pipelines and on-call, about 0.25 FTE.	Highest: both sets of work plus a VPN, two monitoring stacks and two security reviews.
Best fit for my system?	No. Only if a regulator or partner demands on-site data.	Yes. Matches the pilot budget, skills and spiky demand.	Fallback if the tipping point occurs.

Appendix C — Service map and three-year cost comparison

Table C1: Service Map for AWS

Building block	Named service	Model	Container / serverless	Why
Training compute	Amazon SageMaker AI training jobs (ml.m5.xlarge)	PaaS	Managed container	Runs my scikit-learn script weekly (about 250 hours a year); billed only while running; nothing to patch.
Serving / hosting	Amazon ECS on AWS Fargate, behind an Application Load Balancer (Elastic Load Balancing); image in Amazon ECR	PaaS	Container (serverless compute)	Warm tasks across two zones for sub-100 ms scoring; a portable Docker image (Section 3.3).
Data storage	Amazon S3 (server-side encryption with AWS KMS)	IaaS	n/a (storage)	Durable storage for transactions, features, model files and prediction logs; I control keys, lifecycle and access.
Managed AI / ML	Amazon SageMaker AI: Model Registry, Model Monitor, Clarify	PaaS	n/a	Versioning with rollback, drift checks, and SHAP-based reason codes without building the tooling.
Additional: alerting	Amazon SQS, AWS Lambda, Amazon SNS	PaaS	Serverless	About 1% of transactions are flagged, in bursts, each handled in under a second; pay per request.
Additional: network and security	Amazon VPC, AWS Site-to-Site VPN, AWS IAM, AWS KMS, AWS CloudTrail, Amazon CloudWatch	IaaS (VPC, VPN); PaaS (others)	n/a	Private link to the payments platform, least-privilege access, keys, audit log and alarms.

Table C2: Three-year cost comparison (order of magnitude)

Cost line	Assumption and source (with date)	Option A: AWS cloud	Option B: on-premises	Uncertainty (H/M/L)
Compute	A: 2 Fargate tasks (1 vCPU, 2 GB) always on at \$0.04048 per vCPU-hour and \$0.004445 per GB-hour (AWS, n.d.-b), plus 4 extra tasks for 100 hours a year; Application Load Balancer \$0.0225 per hour plus 1.5 capacity units at \$0.008 (AWS, n.d.-d); training 250 hours a year and monitoring 182 hours a year at \$0.23 per hour (CloudBurn, n.d.); all +10% for Canada (assumption). B: 2 CPU servers at \$18k, \$14k network, security and power protection, \$12k a year hosting (my assumptions; confirm with a vendor quote).	\$4.2k	\$86k	M
Storage and data transfer	A: S3 Standard Canada (Central) \$0.025 per GB-month (Enterprise Storage Forum, n.d.); 100, 300 and 600 GB in years 1 to 3; \$100 a year transfer. B: \$5k a year backup storage and network.	\$0.6k	\$15k	L
Managed services / licences	A: per year: CloudWatch \$0.5k, security services \$0.7k, AWS support \$1.2k, queue and alert services \$0.1k (my estimates), Site-to-Site VPN \$0.05 per hour +10% (AWS, n.d.-c). B: OS, virtualization, monitoring and support \$8k a year	\$9.0k	\$24k	M

	(assumption).			
Staff time to operate	\$120k a year loaded cost per FTE (assumption, no source). A: 0.25 FTE. B: 0.75 FTE.	\$90k	\$270k	H
Three- year total	Cloud staff doubled to 0.5 FTE: about \$194k. On- premises with 0.25 FTE: about \$215k. Week 2 model at 50,000 a day: \$88.6k against \$281k.	\$104k	\$395k	